

TEJAS SACHDEVA

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EDUCATION

The University of Texas at Austin – Austin, TX

Master of Science in Mechanical Engineering

May 2027

Bachelor of Science in Mechanical Engineering [GPA: 3.8]

May 2026

EXPERIENCE

Biotex, Inc. – Houston, TX

May 2026 – Aug 2026

Mechanical Engineering Intern

- Designed CNC-machined 6061-T6 assembly fixtures in SolidWorks for subassemblies in a non-invasive glioblastoma therapy helmet, applying DFMA, GD&T, and tolerance stack-ups to prevent press-fit and alignment failure modes across 500+ units
- Redesigned the locking dial and head cradle of the helmet's retention system, conforming the cradle to a 95th-percentile head model; MJF-printed in PA11 (dial) for cyclical fatigue resistance and BASF TPU (cradle) for biocompatibility
- Designed a CNC-machined fixture in Solidworks to meet strict vibration limits for an incubator-based cell culture testing jig; selected Delrin for its non-magnetic and sterilizable properties

Texas Robotics – Austin, TX

Dec 2025 – Present

Research Assistant

- Engineered 3D-printed carbon fiber test fixtures for an upper-limb exoskeleton test bench, capturing human arm-bending kinematics via standardized joint alignment
- Designed the fixtures' load-bearing components to meet a 200 Nm/rad impedance boundary under dynamic loading conditions

General Motors – Warren, MI

May 2025 – Aug 2025

Service Release Engineering Intern (Chassis, Body Exterior)

- Identified part commonality across 2023–2025 Equinox and Terrain platforms, driving \$700K+ in cost savings through BOM simplification and manufacturing alignment
- Analyzed part geometry and critical design features in Siemens Teamcenter to identify common components across vehicle platforms
- Partnered with product and supply chain teams to eliminate duplicate components, reducing service lifecycle complexity

Longhorn Racing Solar – Austin, TX

Aug 2024 – Aug 2025

Powertrain Engineer

- Designed battery cooling manifolds in SolidWorks and ran ANSYS Fluent thermal simulations, achieving 25% improvement in temperature uniformity and 35% reduction in pressure drop under race conditions

Parker Wellbore – Houston, TX

May 2024 – Aug 2024

Technical Services Intern

- Designed a specialized crane basket to safely transport BOP stacks on Parker Wellbore rigs, building 3D models and technical drawings in SolidWorks
- Performed FEA (linear static, modal, fatigue) per API, ASME, and AISC standards to determine FOS and WLL; selected materials via Charpy V-notch testing, achieving a 65% reduction in Von-Mises stress fluctuation under impact loading

LEADERSHIP AND COMMUNITY INVOLVEMENT

First-Year Experience (UT Austin) – Austin, TX

May 2024 – Jan 2025

First-Year Interest Group (FIG) Mentor

- Mentored a cohort of 20+ first-year students through academic and social transition, delivering individualized guidance recognized by mentees as key to their successful adjustment to college

ASME Purdue University Chapter – West Lafayette, IN (prior to transferring to UT Austin)

Oct 2022 – May 2023

Treasurer

- Managed \$20K+ in funds and secured external sponsorships to support engineering competitions and initiatives

ADDITIONAL

- Technical /Computer Skills:** SolidWorks, Autodesk Fusion 360, AutoCAD Mechanical, GD&T, DFMA, Tolerance Analysis, Siemens Teamcenter, ANSYS (Fluent, Mechanical), MATLAB, Python, LabVIEW
- Honors:** Richard Douglas and Judith Watson Perkins Endowed Presidential Scholar
- Certifications:** Design for Six Sigma (Green Belt)
- Patents:** Multi-Spectral Retinal Imaging & Virtual Spectral Decomposition, U.S. Patent App. No. 19/553,050